QA / MANUAL + SOURCE TEST PLAN
Kleym prize distribution
Comprehensive Testnet qualification v1

118 detailed cases | D1-D4 reference approach | Prepared 8 October 2026

PLAN, NOT EXECUTION RESULTS. All new cases are Not Run. Freeze final PR candidate before running. Railway remains paused.

Coverage: D1: 48, D2: 28, D3: 25, D4: 17

Source base: 4163d7954bff9c1582c0c405174cddbe6951da33
Kleym source; Lantern reference packaged artifacts only

0. Scope and qualification boundary

This is a detailed plan for the current Kleym source candidate, not a completed full-wallet or Mainnet test report. Classic Stellar claimable balances, sponsor-funded XLM fees/reserves and sender-owned USDC are under test. Lantern plans provide test discipline, not their contract IDs, scanner risk rules or registry assumptions. Source access is available for Kleym; Lantern artifacts remain packaged builds.
Freeze the final feature commit and deployment identity before execution. Initial cases are Not Run. Local development proof is listed separately; it cannot qualify a different deployed build. Railway remains paused. No service restart, wallet approval, asset movement or Lantern rebuild is authorized by generating this plan.

1. Entry gates and exact environment

G0: record feature commit, clean/dirty source state, lockfile hash, build mode, generated client version, runtime versions and manifest. Halt on unknown candidate.
G1: use disposable isolated migrated PostgreSQL and Redis, restorable fixtures, safe credential injection and reachable approved Horizon. Current local Docker/DB environment is unavailable; infrastructure execution is gated.
G2: owner approves small Testnet balances and signs privately; official Testnet USDC issuer is GBBD47IF6LWK7P7MDEVSCWR7DPUWV3NY3DTQEVFL4NAT4AQH3ZLLFLA5; passphrase Test SDF Network ; September 2015; Horizon https://horizon-testnet.stellar.org. Verify live provider identity before each write. Testnet resets require new snapshots, not recycled historical passes.
G3: approved public HTTPS candidate, exact browser/device/wallet versions and WalletConnect origin configured. Public host previously used: https://kleym-web-testnet.up.railway.app, currently paused; this plan does not assert availability.
G3-Lantern: senior supplies rebuilt curated Apps entry with session=true for exact Kleym origin, verified installed extension ID and Android https://android.golantern.xyz origin. Packaged 0.6.2 does not list Kleym. Generic URL sandbox is not a substitute.

2. Fixtures, budgets and lifecycle matrix

Provision two separate sender modes: literal zero XLM with sponsored account+official USDC trustline, and funded/nonzero XLM. Each must own enough test USDC. Use three recipient modes: absent account, existing zero XLM without USDC trustline, existing zero XLM with official USDC trustline. Use six independent small campaigns to cover all sender/recipient combinations; do not mutate a shared wallet while another case depends on its before state.
Primary tutorial fixture: three links x 1 USDC; A desktop claim, B Android claim, C unused until expiry reclaim. Final net sender debit 2 USDC, two distinct recipients gain 1 each, third balance removed only after sender reclaim; sponsor pays charged fees and reserves. Keep A/B separate recipients for clear snapshots.
Custom prizes fixture: 20,15,10,5,5 = 55 USDC. Capacity fixture: 200 x 1 = 200 USDC, only after operator approves operational budget. Pure int64 extreme amounts never go on-chain. Frozen trustlines use a disposable custom issuer fixture, never tamper with Circle USDC.
Always record base reserve from ledger. Claimable balance with two claimants costs 2 reserve units. Temporary claim account costs 2 plus trustline 1; recipient persistent additions: absent account 3, existing no-trustline 1, already-trusting 0. Sponsor numSponsoring includes unrelated liabilities; use differences and explain retained entries. Fee bid is an upper offer, actual charged fee comes from Horizon. Account spendable = native minus minimum reserve and native selling liabilities.

3. First guided run: three links, three outcomes

1. Pass G0-G3, privately select the intended Testnet sender and record public before balances. Set count 3/amount 1 and a near-future custom expiry that allows both claims.
2. Review exact total, official asset, sponsor reserve/fee bid and expiry UTC. Generate locally and privately save phrase/export. Give no keys, full CSV or claim URLs to logs/chat.
3. Sign exactly the reviewed funding batch; wait for all three funded balances and independent ledger confirmation before sharing. Save public hash/IDs only.
4. Open A privately on desktop, connect recipient then deliberately claim. Record materialisation ledger and strictly later claim ledger, exact USDC credit, zero-XLM state and temporary account removal.
5. Open B privately on Android Chrome/Freighter. Connection does not claim. Explicitly review and approve Claim in wallet; return for confirmed receipt. Lantern repeats require curated rebuilt package first.
6. Leave C unused. After actual on-chain expiry, sender approves reclaim returning exactly 1 USDC. Independently verify recipient balances unchanged and two claimant reserve units released.
7. Refresh campaign, verify statuses and signed receipts. Retry only exact unresolved transaction hashes; never repeat successful funding/claim to obtain a screenshot. Preserve private backups before closing tabs.

4. Execution order and evidence discipline

Run G0 identity and secret-handling checks first; then pure D1/D2 fixtures; real isolated DB/concurrency/recovery; small wallet D1/D3 lifecycle; custom-prize and 200-link performance/capacity; D4 operational qualification. Re-run affected lifecycle and fault cases after code changes. Split source fixture proof from physical device and independently confirmed ledger proof.
Case IDs KLM-D1..D4-AREA-NNN are stable. Status: Not Run -> Ongoing -> Passed/Failed/Blocked; Failed -> For Re-test -> Passed/Failed. Blocked means a named missing prerequisite/proof; record whether infrastructure, dependency, product gap or tester access. N/A requires owner-approved scope reason, never a substitute for missing proof.
For every execution record candidate, UTC time, tester, expected, observed, evidence and defect when Failed. Store files under evidences/runs/<run-id>/<case-id>/; filenames <case-id>_<UTC>_<sequence>. Capture before/after JSON, exact public transaction hashes/ledgers, decoded operations and database journal states. Screenshot explorer/UI with private address bar/phrases/keys excluded. HAR/traces may contain bearer request data: use allowlisted metadata capture or sanitize offline before sharing.
Defect report must name reproducible steps, concrete impact, exact candidate/environment, expected versus actual, severity and priority, source/evidence location and known dependencies. Do not label an unreachable environment a product failure. A negative test passes only when it refuses for the intended reason with no unauthorized financial/state effects.

5. Stop conditions and release criteria

Stop immediately on seed/phrase/bearer disclosure, unintended signing/submission, wrong network/issuer, unexplained principal loss, double payout, missing required signature enforcement, invalid source admission, false ledger success or candidate identity mismatch. Capture only safe evidence and keep affected funds isolated.

P0 money, signing, authorization, journal/recovery and release-gate cases must pass on the frozen candidate with required DB/device/ledger evidence. Any Failed/Blocked/Not Run mandatory case prevents release sign-off. P1 usability findings require explicit owner disposition; excluded cases need reason and owner. Performance SLO, caps, capital, incident/refill ownership, RTO/RPO and rollback triggers must be decided and measured.

No blanket Mainnet-ready claim follows from this plan or local green tests. Mainnet is a separate capped pilot requiring verified Testnet coverage, populated migration/restore, egress enforcement, queue/incident rehearsal and explicit owner approval. Kleym operational treasury sponsors XLM; sender treasury controls USDC principal. Hot/cold split is a design guideline, not an implemented automated refill guarantee.

6. Reference adaptation and known gaps

Lantern D1: borrowed independent before/after money and state accounting, exact artifact identity, negative authorization and hard-stop gates. Kleym uses classic transactions; Lantern Soroban registry IDs, admin operations and fees do not apply.

Lantern D2: borrowed deterministic fixture expectations, malformed/tampered payloads, unavailable dependency handling and source/package separation. Kleym signature/hash/asset/network checks define acceptance; scanner verdict policy is not copied.

Lantern D3 build acceptance and extension plan: borrowed installed artifact hashes, supported-platform repeat, pre-sign review/cancel, actual-versus-spec deviation register and physical device proof. Mock parent tests are not actual Lantern signatures.

Lantern D4 v3: borrowed independent cold reviewer, public URL usability, offline/unavailable honesty, privacy/accessibility and row-by-row evidence/coverage. Kleym requires authenticated sender and wallet signatures; D4 demo no-wallet promises are not copied.

Known release gaps: local dependency audit reports 34 advisories including critical/high (qualify with supported pnpm; remediate/assess); unavailable current local PG/Redis qualification; actual curated Lantern rebuild; latest physical Android signing; SDK delayed multibatch behavior (#105); ledger receipt timestamp parity (#110); DNS/socket egress (#107); bearer-email gate (#108); lifecycle caps and runbooks (#103/#113); pilot approval (#115). Variable-amount renewal currently refuses and is not advertised as supported. GitHub issues are closed only against their full acceptance criteria.

D1 / Protocol, campaign terms and ledger accounting

Every case below starts Not Run. Record exact candidate, observed result and evidence before changing status.

KLM-D1-TER-001 / Equal prizes: three 1-USDC links

P1 | Positive | Not Run | Issues: #99,#114

Preconditions	G0-G3 gates passed; disposable Testnet identities; fresh campaign; privately backed-up keys.
Steps	Create a campaign with count 3, amount 1 and a future custom expiry. Review; save export privately. Fund once after approval.
Expected	Declared total is exactly 3.0000000 USDC; three distinct public link keys; only funded links can be shared.
Required evidence	Review screenshot without phrase/keys; redacted CSV index/amount comparison; public registration and ledger JSON.
Harness / mode	Manual + independent Horizon

KLM-D1-TER-002 / Custom prizes: 20, 15, 10 and two default 5

P0 | Positive | Not Run | Issues: #99,#114

Preconditions	G0-G3 gates passed; disposable Testnet identities; fresh campaign; privately backed-up keys.
Steps	Create count 5/default 5. Set rows 1-3 to 20,15,10; leave rows 4-5 unchanged. Review, export and inspect public registration.
Expected	Row order is stable; amounts are 20,15,10,5,5; exact total 55.0000000; labels, CSV and ledger amounts match.
Required evidence	Review screenshot without phrase/keys; redacted CSV index/amount comparison; public registration and ledger JSON.
Harness / mode	Manual + independent Horizon

KLM-D1-TER-003 / Edit custom prizes across pages

P1 | Positive | Not Run | Issues: #99,#114

Preconditions	G0-G3 gates passed; disposable Testnet identities; fresh campaign; privately backed-up keys.
Steps	Create 25 rows/default 5. Override row 1 and row 21. Move next/back through all pages and return to review.
Expected	Overrides persist at their original indices; no duplicate rows or lost edits; displayed total equals exact decimal sum.
Required evidence	Review screenshot without phrase/keys; redacted CSV index/amount comparison; public registration and ledger JSON.
Harness / mode	Manual + independent Horizon

KLM-D1-TER-004 / Shrink and expand the recipient count

P1 | Positive | Not Run | Issues: #99,#114

Preconditions	G0-G3 gates passed; disposable Testnet identities; fresh campaign; privately backed-up keys.
Steps	Set overrides on rows 1 and 12; shrink count to 10; inspect review. Expand back to 12 and inspect again.
Expected	Only current rows affect totals. Removed overrides never silently fund additional rows; new rows use displayed default.
Required evidence	Review screenshot without phrase/keys; redacted CSV index/amount comparison; public registration and ledger JSON.
Harness / mode	Manual + independent Horizon

KLM-D1-TER-005 / Change default with existing overrides

P1 | Positive | Not Run | Issues: #99,#114

Preconditions	G0-G3 gates passed; disposable Testnet identities; fresh campaign; privately backed-up keys.
Steps	Override one row, change default 5 to 7, then review and export the current list.
Expected	Explicit override is preserved; remaining rows reflect current default; review is the exact list sent for registration.
Required evidence	Review screenshot without phrase/keys; redacted CSV index/amount comparison; public registration and ledger JSON.
Harness / mode	Manual + independent Horizon

KLM-D1-AMT-001 / Minimum positive amount

P0 | Boundary | Not Run | Issues: #99,#114

Preconditions	Pure amount fixtures; isolated migrated PostgreSQL required only for DB precision assertion.
Steps	Validate 0.0000001 through client and API schemas; inspect exact normalized strings; attempt no chain write for extreme fixtures.
Expected	Accept exactly one stroop; no rounding to zero.
Required evidence	Validator output and API error; sanitized DB round-trip record when applicable.
Harness / mode	tests/lib/stellar/amount.test.ts; tests/lib/api/prize-registration.test.ts; isolated DB

KLM-D1-AMT-002 / Zero / negative / text / exponent

P0 | Negative | Not Run | Issues: #99,#114

Preconditions	Pure amount fixtures; isolated migrated PostgreSQL required only for DB precision assertion.
Steps	Validate 0; -1; abc; 1e3 through client and API schemas; inspect exact normalized strings; attempt no chain write for extreme fixtures.
Expected	Refuse each with a useful amount error; no registration or wallet request.
Required evidence	Validator output and API error; sanitized DB round-trip record when applicable.
Harness / mode	tests/lib/stellar/amount.test.ts; tests/lib/api/prize-registration.test.ts; isolated DB

KLM-D1-AMT-003 / Eighth decimal place

P0 | Negative | Not Run | Issues: #99,#114

Preconditions	Pure amount fixtures; isolated migrated PostgreSQL required only for DB precision assertion.
Steps	Validate 1.12345678 through client and API schemas; inspect exact normalized strings; attempt no chain write for extreme fixtures.
Expected	Refuse rather than round or truncate.
Required evidence	Validator output and API error; sanitized DB round-trip record when applicable.
Harness / mode	tests/lib/stellar/amount.test.ts; tests/lib/api/prize-registration.test.ts; isolated DB

KLM-D1-AMT-004 / Maximum classic int64 amount

P0 | Boundary | Not Run | Issues: #99,#114

Preconditions	Pure amount fixtures; isolated migrated PostgreSQL required only for DB precision assertion.
Steps	Validate 922337203685.4775807 through client and API schemas; inspect exact normalized strings; attempt no chain write for extreme fixtures.
Expected	Pure validation accepts the exact maximum. Do not attempt funding this boundary fixture.
Required evidence	Validator output and API error; sanitized DB round-trip record when applicable.
Harness / mode	tests/lib/stellar/amount.test.ts; tests/lib/api/prize-registration.test.ts; isolated DB

KLM-D1-AMT-005 / Amount one stroop above int64

P0 | Negative | Not Run | Issues: #99,#114

Preconditions	Pure amount fixtures; isolated migrated PostgreSQL required only for DB precision assertion.
Steps	Validate 922337203685.4775808 through client and API schemas; inspect exact normalized strings; attempt no chain write for extreme fixtures.
Expected	Pure validation rejects overflow before envelope creation.
Required evidence	Validator output and API error; sanitized DB round-trip record when applicable.
Harness / mode	tests/lib/stellar/amount.test.ts; tests/lib/api/prize-registration.test.ts; isolated DB

KLM-D1-AMT-006 / Exact aggregate and persistence precision

P0 | Boundary | Not Run | Issues: #99,#114

Preconditions	Pure amount fixtures; isolated migrated PostgreSQL required only for DB precision assertion.
Steps	Validate 20;15;10;5;5;0.0000001 through client and API schemas; inspect exact normalized strings; attempt no chain write for extreme fixtures.
Expected	Decimal/stroop total is 55.0000001; DB round trip retains all seven decimal places.
Required evidence	Validator output and API error; sanitized DB round-trip record when applicable.
Harness / mode	tests/lib/stellar/amount.test.ts; tests/lib/api/prize-registration.test.ts; isolated DB

KLM-D1-EXP-001 / Relative expiry buttons

P1 | Boundary | Not Run | Issues: #100,#114

Preconditions	Isolated clock fixtures first; live stale-review case uses approved small Testnet campaign.
Steps	Select 1 hour, 24 hours, 7 days and 30 days from a fixed clock; inspect local and UTC summary for each.
Expected	Each deadline is rounded to minute precision without shortening the selected interval; rerenders do not extend it.
Required evidence	Local/UTC screenshots; fixed-clock output; submitted predicate timestamps.
Harness / mode	tests/lib/campaign/expiry.test.ts; manual browser

KLM-D1-EXP-002 / Custom date and time entry

P1 | Boundary | Not Run | Issues: #100,#114

Preconditions	Isolated clock fixtures first; live stale-review case uses approved small Testnet campaign.
Steps	Choose Custom; type a future complete date/time; tab through fields and inspect the review summary.
Expected	Typed date/time works without calendar dragging; visible timezone and UTC conversion agree with submitted expiry.
Required evidence	Local/UTC screenshots; fixed-clock output; submitted predicate timestamps.
Harness / mode	tests/lib/campaign/expiry.test.ts; manual browser

KLM-D1-EXP-003 / Invalid or past deadline

P1 | Negative | Not Run | Issues: #100,#114

Preconditions	Isolated clock fixtures first; live stale-review case uses approved small Testnet campaign.
Steps	Try empty, partial, impossible calendar date, current instant and past time; continue after each.
Expected	Refuse each invalid/past expiry with a useful inline message; no keys generated or funding prepared.
Required evidence	Local/UTC screenshots; fixed-clock output; submitted predicate timestamps.
Harness / mode	tests/lib/campaign/expiry.test.ts; manual browser

KLM-D1-EXP-004 / Timezone and DST boundaries

P1 | Boundary | Not Run | Issues: #100,#114

Preconditions	Isolated clock fixtures first; live stale-review case uses approved small Testnet campaign.
Steps	Run browser fixtures in Asia/Manila and America/New_York; test DST skipped and repeated local times.
Expected	Skipped local times are refused; chosen offset on a repeated hour is preserved; server and reviewed predicate agree.
Required evidence	Local/UTC screenshots; fixed-clock output; submitted predicate timestamps.
Harness / mode	tests/lib/campaign/expiry.test.ts; manual browser

KLM-D1-EXP-005 / Expiry while review stays open

P1 | Negative | Not Run | Issues: #100,#114

Preconditions	Isolated clock fixtures first; live stale-review case uses approved small Testnet campaign.
Steps	Hold a saved draft until its expiry, then try prepare/sign without changing the reviewed deadline.
Expected	Reject stale expiry; never silently extend a previously reviewed balance predicate.
Required evidence	Local/UTC screenshots; fixed-clock output; submitted predicate timestamps.
Harness / mode	tests/lib/campaign/expiry.test.ts; manual browser

KLM-D1-KEY-001 / Private export is a funding gate

P0 | Security | Not Run | Issues: #108,#114

Preconditions	G0-G3 gates passed; disposable Testnet identities; fresh campaign; privately backed-up keys.
Steps	Generate keys; try sign before saving export. Save it privately and retry.
Expected	Funding stays blocked before export acknowledgment; save unlocks review/sign; no secret reaches the server.
Required evidence	Screenshots with private regions excluded; allowlisted request fields; offline public-key comparison.
Harness / mode	e2e/csv-gate.spec.ts; tests/lib/campaign/derive.test.ts; manual

KLM-D1-KEY-002 / Recovery phrase normalization

P0 | Security | Not Run | Issues: #108,#114

Preconditions	G0-G3 gates passed; disposable Testnet identities; fresh campaign; privately backed-up keys.
Steps	Privately generate a disposable phrase, then paste it as words and as a numbered list in a new draft; compare public key derivation.
Expected	Valid supported BIP39 phrase normalizes deterministically; invalid checksum refuses. Never use a Freighter wallet phrase as test data.
Required evidence	Screenshots with private regions excluded; allowlisted request fields; offline public-key comparison.
Harness / mode	e2e/csv-gate.spec.ts; tests/lib/campaign/derive.test.ts; manual

KLM-D1-KEY-003 / Lost export recovery from saved phrase

P0 | Security | Not Run | Issues: #108,#114

Preconditions	G0-G3 gates passed; disposable Testnet identities; fresh campaign; privately backed-up keys.
Steps	In a clean tab restore the campaign phrase and original derivation index/count; compare public keys offline.
Expected	Same phrase/index recreates the same keys; wrong phrase/index does not. Kleym cannot recover missing private material.
Required evidence	Screenshots with private regions excluded; allowlisted request fields; offline public-key comparison.
Harness / mode	e2e/csv-gate.spec.ts; tests/lib/campaign/derive.test.ts; manual

KLM-D1-KEY-004 / Fragment removal and data minimization

P0 | Security | Not Run | Issues: #108,#114

Preconditions	G0-G3 gates passed; disposable Testnet identities; fresh campaign; privately backed-up keys.
Steps	Open a disposable private claim locally; inspect address bar and sanitized request body after initialization.
Expected	Fragment is cleared from visible history; only public key and signatures/intended public fields leave the browser; no full bearer URL in evidence.
Required evidence	Screenshots with private regions excluded; allowlisted request fields; offline public-key comparison.
Harness / mode	e2e/csv-gate.spec.ts; tests/lib/campaign/derive.test.ts; manual

KLM-D1-MAT-001 / USDC lifecycle: literal-zero-XLM sponsored sender; absent account

P0 | Positive | Not Run | Issues: #97,#98,#99,#100,#101

Preconditions	G0-G3 gates passed; disposable Testnet identities; fresh campaign; privately backed-up keys. Provision literal-zero-XLM sponsored sender and absent account; reserve/capacity snapshots recorded.
Steps	Create two distinct 1-USDC links: one claim and one expiry reclaim. Fund; inspect both balances. Materialise first link; wait for a strictly later ledger; explicitly claim. After the second expiry, sender approves reclaim.
Expected	Sender supplies exactly 2 USDC, receives exactly 1 back; recipient gains exactly 1. Sponsor pays actual fees/reserves. Zero-XLM actors remain zero. Temporary account/trustline removed, persistent recipient entries correctly sponsored.
Required evidence	Public before/after Horizon account JSON, transaction and operation JSON, prepared-XDR hash, DB rows with bearer material excluded. Include materialisation hash and ledger, claim hash and later ledger, reclaim hash; IDs removed independently.
Harness / mode	scripts/verify-sponsored-funding.ts; scripts/verify-app-lifecycle.ts plus fresh supported-wallet approvals

KLM-D1-MAT-002 / USDC lifecycle: literal-zero-XLM sponsored sender; existing zero-XLM account without USDC trustline

P0 | Positive | Not Run | Issues: #97,#98,#99,#100,#101

Preconditions	G0-G3 gates passed; disposable Testnet identities; fresh campaign; privately backed-up keys. Provision literal-zero-XLM sponsored sender and existing zero-XLM account without USDC trustline; reserve/capacity snapshots recorded.
Steps	Create two distinct 1-USDC links: one claim and one expiry reclaim. Fund; inspect both balances. Materialise first link; wait for a strictly later ledger; explicitly claim. After the second expiry, sender approves reclaim.
Expected	Sender supplies exactly 2 USDC, receives exactly 1 back; recipient gains exactly 1. Sponsor pays actual fees/reserves. Zero-XLM actors remain zero. Temporary account/trustline removed, persistent recipient entries correctly sponsored.
Required evidence	Public before/after Horizon account JSON, transaction and operation JSON, prepared-XDR hash, DB rows with bearer material excluded. Include materialisation hash and ledger, claim hash and later ledger, reclaim hash; IDs removed independently.
Harness / mode	scripts/verify-sponsored-funding.ts; scripts/verify-app-lifecycle.ts plus fresh supported-wallet approvals

KLM-D1-MAT-003 / USDC lifecycle: literal-zero-XLM sponsored sender; existing zero-XLM account with USDC trustline

P0 | Positive | Not Run | Issues: #97,#98,#99,#100,#101

Preconditions	G0-G3 gates passed; disposable Testnet identities; fresh campaign; privately backed-up keys. Provision literal-zero-XLM sponsored sender and existing zero-XLM account with USDC trustline; reserve/capacity snapshots recorded.
Steps	Create two distinct 1-USDC links: one claim and one expiry reclaim. Fund; inspect both balances. Materialise first link; wait for a strictly later ledger; explicitly claim. After the second expiry, sender approves reclaim.
Expected	Sender supplies exactly 2 USDC, receives exactly 1 back; recipient gains exactly 1. Sponsor pays actual fees/reserves. Zero-XLM actors remain zero. Temporary account/trustline removed, persistent recipient entries correctly sponsored.
Required evidence	Public before/after Horizon account JSON, transaction and operation JSON, prepared-XDR hash, DB rows with bearer material excluded. Include materialisation hash and ledger, claim hash and later ledger, reclaim hash; IDs removed independently.
Harness / mode	scripts/verify-sponsored-funding.ts; scripts/verify-app-lifecycle.ts plus fresh supported-wallet approvals

KLM-D1-MAT-004 / USDC lifecycle: funded sender with nonzero XLM; absent account

P0 | Positive | Not Run | Issues: #97,#98,#99,#100,#101

Preconditions	G0-G3 gates passed; disposable Testnet identities; fresh campaign; privately backed-up keys. Provision funded sender with nonzero XLM and absent account; reserve/capacity snapshots recorded.
Steps	Create two distinct 1-USDC links: one claim and one expiry reclaim. Fund; inspect both balances. Materialise first link; wait for a strictly later ledger; explicitly claim. After the second expiry, sender approves reclaim.
Expected	Sender supplies exactly 2 USDC, receives exactly 1 back; recipient gains exactly 1. Sponsor pays actual fees/reserves. Zero-XLM actors remain zero. Temporary account/trustline removed, persistent recipient entries correctly sponsored.
Required evidence	Public before/after Horizon account JSON, transaction and operation JSON, prepared-XDR hash, DB rows with bearer material excluded. Include materialisation hash and ledger, claim hash and later ledger, reclaim hash; IDs removed independently.
Harness / mode	scripts/verify-sponsored-funding.ts; scripts/verify-app-lifecycle.ts plus fresh supported-wallet approvals

KLM-D1-MAT-005 / USDC lifecycle: funded sender with nonzero XLM; existing zero-XLM account without USDC trustline

P0 | Positive | Not Run | Issues: #97,#98,#99,#100,#101

Preconditions	G0-G3 gates passed; disposable Testnet identities; fresh campaign; privately backed-up keys. Provision funded sender with nonzero XLM and existing zero-XLM account without USDC trustline; reserve/capacity snapshots recorded.
Steps	Create two distinct 1-USDC links: one claim and one expiry reclaim. Fund; inspect both balances. Materialise first link; wait for a strictly later ledger; explicitly claim. After the second expiry, sender approves reclaim.
Expected	Sender supplies exactly 2 USDC, receives exactly 1 back; recipient gains exactly 1. Sponsor pays actual fees/reserves. Zero-XLM actors remain zero. Temporary account/trustline removed, persistent recipient entries correctly sponsored.
Required evidence	Public before/after Horizon account JSON, transaction and operation JSON, prepared-XDR hash, DB rows with bearer material excluded. Include materialisation hash and ledger, claim hash and later ledger, reclaim hash; IDs removed independently.
Harness / mode	scripts/verify-sponsored-funding.ts; scripts/verify-app-lifecycle.ts plus fresh supported-wallet approvals

KLM-D1-MAT-006 / USDC lifecycle: funded sender with nonzero XLM; existing zero-XLM account with USDC trustline

P0 | Positive | Not Run | Issues: #97,#98,#99,#100,#101

Preconditions	G0-G3 gates passed; disposable Testnet identities; fresh campaign; privately backed-up keys. Provision funded sender with nonzero XLM and existing zero-XLM account with USDC trustline; reserve/capacity snapshots recorded.
Steps	Create two distinct 1-USDC links: one claim and one expiry reclaim. Fund; inspect both balances. Materialise first link; wait for a strictly later ledger; explicitly claim. After the second expiry, sender approves reclaim.
Expected	Sender supplies exactly 2 USDC, receives exactly 1 back; recipient gains exactly 1. Sponsor pays actual fees/reserves. Zero-XLM actors remain zero. Temporary account/trustline removed, persistent recipient entries correctly sponsored.
Required evidence	Public before/after Horizon account JSON, transaction and operation JSON, prepared-XDR hash, DB rows with bearer material excluded. Include materialisation hash and ledger, claim hash and later ledger, reclaim hash; IDs removed independently.
Harness / mode	scripts/verify-sponsored-funding.ts; scripts/verify-app-lifecycle.ts plus fresh supported-wallet approvals

KLM-D1-FUN-001 / Funding envelope ownership

P0 | Regression | Not Run | Issues: #98,#99,#103,#105

Preconditions	G0-G3 gates passed; disposable Testnet identities; fresh campaign; privately backed-up keys.
Steps	Prepare a fresh 3-link USDC campaign; decode operations, source accounts, signers, asset and predicates before approving.
Expected	Classic source is operational sponsor; sender supplies principal in createClaimableBalance ops; sponsoring brackets wrap funding; no fee bump substitutes for reserve sponsorship.
Required evidence	Public before/after Horizon account JSON, transaction and operation JSON, prepared-XDR hash, DB rows with bearer material excluded.
Harness / mode	tests/lib/api/prize-funding.test.ts; verify-sponsored-funding; verify-batches; wallet

KLM-D1-FUN-002 / Funding signer rejection

P0 | Negative | Not Run | Issues: #98,#99,#103,#105

Preconditions	G0-G3 gates passed; disposable Testnet identities; fresh campaign; privately backed-up keys.
Steps	Remove sender signature, then sponsor signature from disposable prepared fixture; submit to isolated API.
Expected	Missing/wrong required signature is refused before network submission; pending journal cannot mark funding confirmed.
Required evidence	Public before/after Horizon account JSON, transaction and operation JSON, prepared-XDR hash, DB rows with bearer material excluded.
Harness / mode	tests/lib/api/prize-funding.test.ts; verify-sponsored-funding; verify-batches; wallet

KLM-D1-FUN-003 / Balance ID derivation and reserve units

P0 | Regression | Not Run | Issues: #98,#99,#103,#105

Preconditions	G0-G3 gates passed; disposable Testnet identities; fresh campaign; privately backed-up keys.
Steps	Fund 3 links once; compare stored IDs against createClaimableBalance operation order and Horizon.
Expected	Three exact asset/amount/claimant predicates and balance IDs match; sponsor adds 6 reserve units for two claimants per balance.
Required evidence	Public before/after Horizon account JSON, transaction and operation JSON, prepared-XDR hash, DB rows with bearer material excluded.
Harness / mode	tests/lib/api/prize-funding.test.ts; verify-sponsored-funding; verify-batches; wallet

KLM-D1-FUN-004 / 200 links across batches

P0 | Regression | Not Run | Issues: #98,#99,#103,#105

Preconditions	G0-G3 gates passed; disposable Testnet identities; fresh campaign; privately backed-up keys.
Steps	With approved Testnet budget, create 200 one-USDC links; prepare and sign sequentially using configured batch size; wait for every batch confirmation.
Expected	All 200 IDs unique and confirmed exactly once. Number/signature count follows effective per-envelope operation cap; no invalid >100-op envelope; campaign active only after all required confirmations.
Required evidence	Public before/after Horizon account JSON, transaction and operation JSON, prepared-XDR hash, DB rows with bearer material excluded.
Harness / mode	tests/lib/api/prize-funding.test.ts; verify-sponsored-funding; verify-batches; wallet

KLM-D1-FUN-005 / Principal shortage

P0 | Negative | Not Run | Issues: #98,#99,#103,#105

Preconditions	G0-G3 gates passed; disposable Testnet identities; fresh campaign; privately backed-up keys.
Steps	Use sender with 2 USDC and request three 1-USDC links; try funding.
Expected	Fresh preflight refuses insufficient USDC before wallet signature; no link marked funded or active.
Required evidence	Public before/after Horizon account JSON, transaction and operation JSON, prepared-XDR hash, DB rows with bearer material excluded.
Harness / mode	tests/lib/api/prize-funding.test.ts; verify-sponsored-funding; verify-batches; wallet

KLM-D1-FUN-006 / Sponsor shortage / liabilities

P0 | Negative | Not Run | Issues: #98,#99,#103,#105

Preconditions	G0-G3 gates passed; disposable Testnet identities; fresh campaign; privately backed-up keys.
Steps	Use isolated sponsor fixtures with low spendable XLM and native selling liabilities; prepare full remaining campaign.
Expected	Spendable uses fresh base reserve, entries, sponsorship and selling liabilities; insufficient full remaining capacity refuses before signing.
Required evidence	Public before/after Horizon account JSON, transaction and operation JSON, prepared-XDR hash, DB rows with bearer material excluded.
Harness / mode	tests/lib/api/prize-funding.test.ts; verify-sponsored-funding; verify-batches; wallet

KLM-D1-FUN-007 / Account clone and sequence stability

P0 | Regression | Not Run | Issues: #98,#99,#103,#105

Preconditions	G0-G3 gates passed; disposable Testnet identities; fresh campaign; privately backed-up keys.
Steps	Prepare twice from shared account fixture and compare originals; then perform actual sequential confirmed batches in isolated harness.
Expected	SDK builder does not mutate cached sender/sponsor account records; each accepted envelope uses coordinated correct source sequence.
Required evidence	Public before/after Horizon account JSON, transaction and operation JSON, prepared-XDR hash, DB rows with bearer material excluded.
Harness / mode	tests/lib/api/prize-funding.test.ts; verify-sponsored-funding; verify-batches; wallet

KLM-D1-FUN-008 / Legacy native-XLM compatibility

P0 | Regression | Not Run | Issues: #98,#99,#103,#105

Preconditions	G0-G3 gates passed; disposable Testnet identities; fresh campaign; privately backed-up keys.
Steps	Create a small XLM campaign using explicitly approved test identity with spendable XLM; fund, claim and reclaim unused balance.
Expected	Legacy native funding path remains compatible; exact native principal, fees and reserves reconcile; sponsor policy follows reviewed envelope.
Required evidence	Public before/after Horizon account JSON, transaction and operation JSON, prepared-XDR hash, DB rows with bearer material excluded.
Harness / mode	tests/lib/api/prize-funding.test.ts; verify-sponsored-funding; verify-batches; wallet

KLM-D1-SET-001 / Claim exactly once

P0 | Regression | Not Run | Issues: #100,#101,#110,#114

Preconditions	G0-G3 gates passed; disposable Testnet identities; fresh campaign; privately backed-up keys.
Steps	For a funded link, connect recipient; record no mutation on connect; claim once and reload/click again.
Expected	Only explicit claim moves funds; exact amount credited once; claimed screen shows confirmed hash; repeated claim cannot pay again.
Required evidence	Public before/after Horizon account JSON, transaction and operation JSON, prepared-XDR hash, DB rows with bearer material excluded. Sanitized receipt canonical bytes/signature/public verification key when relevant.
Harness / mode	Manual + independent Horizon

KLM-D1-SET-002 / One-stroop custom claim

P0 | Regression | Not Run | Issues: #100,#101,#110,#114

Preconditions	G0-G3 gates passed; disposable Testnet identities; fresh campaign; privately backed-up keys.
Steps	Fund a custom 0.0000001-USDC link after approval; claim to trusted recipient.
Expected	Recipient gets exactly 0.0000001; totals and receipt preserve seven decimals; no zero rounding.
Required evidence	Public before/after Horizon account JSON, transaction and operation JSON, prepared-XDR hash, DB rows with bearer material excluded. Sanitized receipt canonical bytes/signature/public verification key when relevant.
Harness / mode	Manual + independent Horizon

KLM-D1-SET-003 / Frozen / unauthorized / full trustline

P0 | Negative | Not Run | Issues: #100,#101,#110,#114

Preconditions	G0-G3 gates passed; disposable Testnet identities; fresh campaign; privately backed-up keys.
Steps	Use disposable issuer fixture with unauthorized or full receiving trustline; try prepare and claim.
Expected	Refuse unusable destination without losing balance or misreporting success; Circle USDC must not be altered for this test.
Required evidence	Public before/after Horizon account JSON, transaction and operation JSON, prepared-XDR hash, DB rows with bearer material excluded. Sanitized receipt canonical bytes/signature/public verification key when relevant.
Harness / mode	Manual + independent Horizon

KLM-D1-SET-004 / Claim before / at / after expiry

P0 | Regression | Not Run | Issues: #100,#101,#110,#114

Preconditions	G0-G3 gates passed; disposable Testnet identities; fresh campaign; privately backed-up keys.
Steps	Using controlled ledger close-time fixture, test T-1,T,T+1. In approved live boundary run, compare actual Horizon predicate seconds and ledger close time.
Expected	Ephemeral claimant eligible only abs_before T; sender reclaim eligible at/after T. Host clock and millisecond UI cannot redefine predicate.
Required evidence	Public before/after Horizon account JSON, transaction and operation JSON, prepared-XDR hash, DB rows with bearer material excluded. Sanitized receipt canonical bytes/signature/public verification key when relevant.
Harness / mode	Manual + independent Horizon

KLM-D1-SET-005 / Temporary reserve release

P0 | Regression | Not Run | Issues: #100,#101,#110,#114

Preconditions	G0-G3 gates passed; disposable Testnet identities; fresh campaign; privately backed-up keys.
Steps	Record sponsor before materialisation, after materialisation and after claim. Independently check temporary account absence and removed balance.
Expected	Temporary account 2 plus trustline 1 reserve units released; claimable-balance 2 units released; permanent recipient sponsored entries remain and are separately explained.
Required evidence	Public before/after Horizon account JSON, transaction and operation JSON, prepared-XDR hash, DB rows with bearer material excluded. Sanitized receipt canonical bytes/signature/public verification key when relevant.
Harness / mode	Manual + independent Horizon

KLM-D1-SET-006 / Sender-authorized reclaim after expiry

P0 | Regression | Not Run | Issues: #100,#101,#110,#114

Preconditions	G0-G3 gates passed; disposable Testnet identities; fresh campaign; privately backed-up keys.
Steps	Keep third link unused. Wait for on-chain expiry; select sender; review single reclaim amount/signers; approve once.
Expected	Unused 1 USDC returns only to original sender; balance disappears; 2 claimant reserve units released; sponsor pays actual fee; recipient state unchanged.
Required evidence	Public before/after Horizon account JSON, transaction and operation JSON, prepared-XDR hash, DB rows with bearer material excluded. Sanitized receipt canonical bytes/signature/public verification key when relevant.
Harness / mode	Manual + independent Horizon

KLM-D1-SET-007 / Pre-expiry reclaim / wrong sender

P0 | Negative | Not Run | Issues: #100,#101,#110,#114

Preconditions	G0-G3 gates passed; disposable Testnet identities; fresh campaign; privately backed-up keys.
Steps	Try reclaim one second before deadline and with another sender identity in controlled API/ledger fixtures.
Expected	Refuse both; no sender refund, no removed balance and no journal settlement from a rejection.
Required evidence	Public before/after Horizon account JSON, transaction and operation JSON, prepared-XDR hash, DB rows with bearer material excluded. Sanitized receipt canonical bytes/signature/public verification key when relevant.
Harness / mode	Manual + independent Horizon

KLM-D1-SET-008 / Partial reclaim and duplicate reconciliation

P0 | Regression | Not Run | Issues: #100,#101,#110,#114

Preconditions	G0-G3 gates passed; disposable Testnet identities; fresh campaign; privately backed-up keys.
Steps	Use 3 expired links with one already claimed; reclaim remaining links; replay worker/event and refresh campaign.
Expected	Only reclaimable amounts are returned; sum/receipt count matches exact settled links; duplicate processing cannot credit twice.
Required evidence	Public before/after Horizon account JSON, transaction and operation JSON, prepared-XDR hash, DB rows with bearer material excluded. Sanitized receipt canonical bytes/signature/public verification key when relevant.
Harness / mode	Manual + independent Horizon

KLM-D1-SET-009 / Renewal uses verified refunded budget

P0 | Regression | Not Run | Issues: #100,#101,#110,#114

Preconditions	G0-G3 gates passed; disposable Testnet identities; fresh campaign; privately backed-up keys.
Steps	Expire an equal-amount campaign, reclaim remaining links, wait for confirmed accounting, then renew from original campaign.
Expected	Original lineage retained; renewed link count and principal equal verified refund only; claimed links are excluded; renewed funding still requires sender review.
Required evidence	Public before/after Horizon account JSON, transaction and operation JSON, prepared-XDR hash, DB rows with bearer material excluded. Sanitized receipt canonical bytes/signature/public verification key when relevant.
Harness / mode	Manual + independent Horizon

KLM-D1-SET-010 / Renewal refuses unsafe terms

P0 | Negative | Not Run | Issues: #100,#101,#110,#114

Preconditions	G0-G3 gates passed; disposable Testnet identities; fresh campaign; privately backed-up keys.
Steps	Try renewal before expiry, before refund settles, with changed sender/asset/count, after second renewal, and on custom-amount campaign.
Expected	Refuse invalid terms; variable-amount renewal is currently unsupported and clearly refused; no second principal distribution hidden as renewal.
Required evidence	Public before/after Horizon account JSON, transaction and operation JSON, prepared-XDR hash, DB rows with bearer material excluded. Sanitized receipt canonical bytes/signature/public verification key when relevant.
Harness / mode	Manual + independent Horizon

KLM-D1-SET-011 / Ledger-derived receipt time

P0 | Regression | Not Run | Issues: #100,#101,#110,#114

Preconditions	G0-G3 gates passed; disposable Testnet identities; fresh campaign; privately backed-up keys.
Steps	Claim, stop UI before DB settles, and run reconciliation; compare direct and recovered receipt timestamps with confirmed operation created_at.
Expected	Both paths use ledger settlement time consistently. Current issue #110 remains a known gap until verified/fixed; app wall-clock time is insufficient.
Required evidence	Public before/after Horizon account JSON, transaction and operation JSON, prepared-XDR hash, DB rows with bearer material excluded. Sanitized receipt canonical bytes/signature/public verification key when relevant.
Harness / mode	Manual + independent Horizon

KLM-D1-POL-001 / Bearer policy versus named recipient

P0 | Regression | Not Run | Issues: #101,#114,#116

Preconditions	G0-G3 gates passed; disposable Testnet identities; fresh campaign; privately backed-up keys. Use isolated policy/conversion fixtures and disposable issuers; real exchange deposits are not authorized.
Steps	Share a disposable bearer link with a different controlled recipient; compare with an explicitly configured strict-policy fixture.
Expected	Bearer policy is not email/address binding; strict policy follows its documented authorization. Never describe sender labels as claimant enforcement.
Required evidence	Public before/after Horizon account JSON, transaction and operation JSON, prepared-XDR hash, DB rows with bearer material excluded. Safe policy result, destination asset/minimum and memo type; no OTP/token capture.
Harness / mode	verify:claim-policies; verify:claims; tests/lib/api/receipts.test.ts; approved compatibility wallet cases

KLM-D1-POL-002 / Receive-as quote and slippage floor

P0 | Regression | Not Run | Issues: #101,#114,#116

Preconditions	G0-G3 gates passed; disposable Testnet identities; fresh campaign; privately backed-up keys. Use isolated policy/conversion fixtures and disposable issuers; real exchange deposits are not authorized.
Steps	In an approved disposable asset pair, quote an allowlisted receive-as asset; change quote liquidity and try below-minimum destination before submission.
Expected	Delivered asset and minimum are explicitly reviewed; failed conversion preserves claimable principal; no silent ticker/issuer substitution.
Required evidence	Public before/after Horizon account JSON, transaction and operation JSON, prepared-XDR hash, DB rows with bearer material excluded. Safe policy result, destination asset/minimum and memo type; no OTP/token capture.
Harness / mode	verify:claim-policies; verify:claims; tests/lib/api/receipts.test.ts; approved compatibility wallet cases

KLM-D1-POL-003 / Exchange deposit memo boundaries

P0 | Regression | Not Run | Issues: #101,#114,#116

Preconditions	G0-G3 gates passed; disposable Testnet identities; fresh campaign; privately backed-up keys. Use isolated policy/conversion fixtures and disposable issuers; real exchange deposits are not authorized.
Steps	Use controlled exchange-style account fixture; test required ID/text memo, missing/wrong-type/overflow memo and disabled exchange-deposit campaign.
Expected	Only permitted reviewed destination/memo is accepted; no recipient account merge; missing or altered memo is refused before transfer; actual delivered asset appears on receipt.
Required evidence	Public before/after Horizon account JSON, transaction and operation JSON, prepared-XDR hash, DB rows with bearer material excluded. Safe policy result, destination asset/minimum and memo type; no OTP/token capture.
Harness / mode	verify:claim-policies; verify:claims; tests/lib/api/receipts.test.ts; approved compatibility wallet cases

D2 / Signing, authorization and recovery under faults

Every case below starts Not Run. Record exact candidate, observed result and evidence before changing status.

KLM-D2-SIG-001 / Network/provider mismatch

P0 | Negative | Not Run | Issues: #96,#98

Preconditions	Isolated source/DB/provider fixtures; network-fault injection is local only; no public denial-of-service.
Steps	Configure Testnet passphrase with public Horizon or vice versa; attempt every write preparation.
Expected	Refuse mismatched network identity before wallet or network write; read-only Mainnet checks cannot submit.
Required evidence	Full sanitized harness output; exact hashes; journal/DB snapshots; process IDs and timestamps.
Harness / mode	tests/lib/wallet/signing.test.ts; tests/lib/stellar/source-admission.test.ts; verify-source-coordination; isolated recovery harness

KLM-D2-SIG-002 / Wrong asset issuer

P0 | Negative | Not Run | Issues: #98

Preconditions	Isolated source/DB/provider fixtures; network-fault injection is local only; no public denial-of-service.
Steps	Alter prepared or requested USDC issuer while keeping display code USDC.
Expected	Match full asset code+issuer+network; refuse unauthorized issuer, never silently map by ticker.
Required evidence	Full sanitized harness output; exact hashes; journal/DB snapshots; process IDs and timestamps.
Harness / mode	tests/lib/wallet/signing.test.ts; tests/lib/stellar/source-admission.test.ts; verify-source-coordination; isolated recovery harness

KLM-D2-SIG-003 / Reviewed XDR tampering

P0 | Security | Not Run | Issues: #98,#101

Preconditions	Isolated source/DB/provider fixtures; network-fault injection is local only; no public denial-of-service.
Steps	Alter amount, destination, source, operation order, predicates, timebounds and fee separately in signing fixtures.
Expected	Every altered reviewed transaction hash is refused; no mutation accepted because a wallet signed it.
Required evidence	Full sanitized harness output; exact hashes; journal/DB snapshots; process IDs and timestamps.
Harness / mode	tests/lib/wallet/signing.test.ts; tests/lib/stellar/source-admission.test.ts; verify-source-coordination; isolated recovery harness

KLM-D2-SIG-004 / Signed envelope preservation

P0 | Security | Not Run | Issues: #98,#101

Preconditions	Isolated source/DB/provider fixtures; network-fault injection is local only; no public denial-of-service.
Steps	Return signed fixtures missing sponsor/local signatures or containing a signature for another passphrase/account.
Expected	Require exact reviewed hash and all required valid signatures; do not reconstruct or drop prior signatures.
Required evidence	Full sanitized harness output; exact hashes; journal/DB snapshots; process IDs and timestamps.
Harness / mode	tests/lib/wallet/signing.test.ts; tests/lib/stellar/source-admission.test.ts; verify-source-coordination; isolated recovery harness

KLM-D2-SIG-005 / Wallet account/network changes mid-review

P0 | Negative | Not Run | Issues: #106

Preconditions	Isolated source/DB/provider fixtures; network-fault injection is local only; no public denial-of-service.
Steps	Connect sender, prepare; switch selected wallet or network before sign. Retry only after fresh review.
Expected	Refuse changed identity; reconnect/review clearly; no automatic retry with different account/network.
Required evidence	Full sanitized harness output; exact hashes; journal/DB snapshots; process IDs and timestamps.
Harness / mode	tests/lib/wallet/signing.test.ts; tests/lib/stellar/source-admission.test.ts; verify-source-coordination; isolated recovery harness

KLM-D2-SIG-006 / Cancel / timeout / lock wallet

P0 | Negative | Not Run | Issues: #106,#114

Preconditions	Isolated source/DB/provider fixtures; network-fault injection is local only; no public denial-of-service.
Steps	Cancel connection and signing separately; lock wallet; let acknowledgment/review timeout; retry from saved draft.
Expected	Useful bounded error; no funding or claim on cancel; cleanup permits safe retry; no leaked wallet error contents.
Required evidence	Full sanitized harness output; exact hashes; journal/DB snapshots; process IDs and timestamps.
Harness / mode	tests/lib/wallet/signing.test.ts; tests/lib/stellar/source-admission.test.ts; verify-source-coordination; isolated recovery harness

KLM-D2-SIG-007 / Unknown submission outcome

P0 | Negative | Not Run | Issues: #104

Preconditions	Isolated source/DB/provider fixtures; network-fault injection is local only; no public denial-of-service.
Steps	Drop response after provider accepts exact signed hash; refresh/retry and run reconciliation.
Expected	Exact signed bytes/hash retained durably; lookup same hash before any retry; no replacement payment or duplicate funding.
Required evidence	Full sanitized harness output; exact hashes; journal/DB snapshots; process IDs and timestamps.
Harness / mode	tests/lib/wallet/signing.test.ts; tests/lib/stellar/source-admission.test.ts; verify-source-coordination; isolated recovery harness

KLM-D2-SIG-008 / Process crash at durable boundaries

P0 | Negative | Not Run | Issues: #104

Preconditions	Isolated source/DB/provider fixtures; network-fault injection is local only; no public denial-of-service.
Steps	Kill isolated worker before send, after send, after confirm and during settlement; restart for each journal fixture.
Expected	Recover prepared/submitted/confirmed/settled state deterministically; exactly one final accounting mutation per hash.
Required evidence	Full sanitized harness output; exact hashes; journal/DB snapshots; process IDs and timestamps.
Harness / mode	tests/lib/wallet/signing.test.ts; tests/lib/stellar/source-admission.test.ts; verify-source-coordination; isolated recovery harness

KLM-D2-SIG-009 / Real source concurrency across processes

P0 | Negative | Not Run | Issues: #102,#103

Preconditions	Isolated source/DB/provider fixtures; network-fault injection is local only; no public denial-of-service.
Steps	Run two processes sharing sponsor and separate web/worker instances against fresh PostgreSQL; prepare claims/funding/reclaim simultaneously.
Expected	PostgreSQL advisory source lock serializes all spenders; fresh sequence/headroom under lock; no stale capacity double admission.
Required evidence	Full sanitized harness output; exact hashes; journal/DB snapshots; process IDs and timestamps.
Harness / mode	tests/lib/wallet/signing.test.ts; tests/lib/stellar/source-admission.test.ts; verify-source-coordination; isolated recovery harness

KLM-D2-SIG-010 / Unresolved legacy attempt

P0 | Negative | Not Run | Issues: #104

Preconditions	Isolated source/DB/provider fixtures; network-fault injection is local only; no public denial-of-service.
Steps	Seed old submitted row with missing signed XDR and ambiguous hash; attempt new prepare and recovery.
Expected	Do not discard unresolved financial liability by guessing; refuse/reconcile explicitly; no duplicate operation.
Required evidence	Full sanitized harness output; exact hashes; journal/DB snapshots; process IDs and timestamps.
Harness / mode	tests/lib/wallet/signing.test.ts; tests/lib/stellar/source-admission.test.ts; verify-source-coordination; isolated recovery harness

KLM-D2-SIG-011 / Strictly later ledger enforcement

P0 | Negative | Not Run | Issues: #101

Preconditions	Isolated source/DB/provider fixtures; network-fault injection is local only; no public denial-of-service.
Steps	Hold TX1 confirmed ledger N; provide provider latest ledger N for TX2 then N+1; attempt claim preparation both times.
Expected	Refuse N; permit after verified N+1. Fixed sleep or browser time alone is insufficient.
Required evidence	Full sanitized harness output; exact hashes; journal/DB snapshots; process IDs and timestamps.
Harness / mode	tests/lib/wallet/signing.test.ts; tests/lib/stellar/source-admission.test.ts; verify-source-coordination; isolated recovery harness

KLM-D2-SIG-012 / Queue/DB unavailable

P0 | Negative | Not Run | Issues: #104,#113

Preconditions	Isolated source/DB/provider fixtures; network-fault injection is local only; no public denial-of-service.
Steps	Interrupt Redis/PG before and after financial write; restore and replay pending jobs.
Expected	No false success; durable attempts survive lost queue notification; recover accounting after reconnection without duplicate payout.
Required evidence	Full sanitized harness output; exact hashes; journal/DB snapshots; process IDs and timestamps.
Harness / mode	tests/lib/wallet/signing.test.ts; tests/lib/stellar/source-admission.test.ts; verify-source-coordination; isolated recovery harness

KLM-D2-SIG-013 / Source rejection classification

P0 | Negative | Not Run | Issues: #104

Preconditions	Isolated source/DB/provider fixtures; network-fault injection is local only; no public denial-of-service.
Steps	Inject tx_bad_seq, insufficient balance, expired timebound and operation failure into isolated submission/reconcile harness.
Expected	Distinguish explicit rejection from unknown outcome; settle nothing on rejection; preserve useful safe correlation metadata.
Required evidence	Full sanitized harness output; exact hashes; journal/DB snapshots; process IDs and timestamps.
Harness / mode	tests/lib/wallet/signing.test.ts; tests/lib/stellar/source-admission.test.ts; verify-source-coordination; isolated recovery harness

KLM-D2-SIG-014 / SDK sequential multi-batch handling

P0 | Negative | Not Run | Issues: #105

Preconditions	Isolated source/DB/provider fixtures; network-fault injection is local only; no public denial-of-service.
Steps	Use published SDK/API client for >one batch; introduce delayed confirmation and exercise resume.
Expected	SDK waits/queries confirmation or surfaces actionable resumable state before next preparation; SUBMITTED is not CONFIRMED. Issue #105 remains open.
Required evidence	Full sanitized harness output; exact hashes; journal/DB snapshots; process IDs and timestamps.
Harness / mode	tests/lib/wallet/signing.test.ts; tests/lib/stellar/source-admission.test.ts; verify-source-coordination; isolated recovery harness

KLM-D2-SEC-001 / Cross-organisation and role controls

P0 | Security | Not Run | Issues: #114

Preconditions	Disposable accounts/fixtures only; local authorized fault injection; owner-approved egress test target for DNS case.
Steps	Create two disposable orgs; try reading/registering/funding/reclaiming other org campaign with session and API key; try read-only role.
Expected	Resource/role denial without secret/object leakage; no rows or chain submissions created for unauthorized request.
Required evidence	Sanitized request/response and DB evidence; guard test logs; no credential values or bearer URLs.
Harness / mode	Existing verify:concurrency, verify:api-keys, verify:privacy, verify:email, source tests; controlled egress check

KLM-D2-SEC-002 / Idempotency scope and request mismatch

P0 | Security | Not Run | Issues: #102,#114

Preconditions	Disposable accounts/fixtures only; local authorized fault injection; owner-approved egress test target for DNS case.
Steps	Reuse same key with same request, changed body and different user/org; run concurrently.
Expected	Same authorized request replays one result; changed body refuses; key never returns another org response.
Required evidence	Sanitized request/response and DB evidence; guard test logs; no credential values or bearer URLs.
Harness / mode	Existing verify:concurrency, verify:api-keys, verify:privacy, verify:email, source tests; controlled egress check

KLM-D2-SEC-003 / API key underscores and secret at rest

P0 | Security | Not Run | Issues: #112

Preconditions	Disposable accounts/fixtures only; local authorized fault injection; owner-approved egress test target for DNS case.
Steps	Use deterministic full API-key fixtures with underscores in suffix; verify authentication, revoke/wrong-key denial and database representation.
Expected	Parser/hash covers whole token; no probabilistic generation assumption or partial split token check; no cleartext key at rest. Issue #112 harness gap must be fixed.
Required evidence	Sanitized request/response and DB evidence; guard test logs; no credential values or bearer URLs.
Harness / mode	Existing verify:concurrency, verify:api-keys, verify:privacy, verify:email, source tests; controlled egress check

KLM-D2-SEC-004 / Claim proof and body guard

P0 | Security | Not Run | Issues: #101,#114

Preconditions	Disposable accounts/fixtures only; local authorized fault injection; owner-approved egress test target for DNS case.
Steps	Submit malformed/oversize body, planted secret field, wrong preparation proof, replayed nonce and altered recipient.
Expected	Refuse at API boundary; no logs reflect bearer material; only valid matching proof advances intended claim.
Required evidence	Sanitized request/response and DB evidence; guard test logs; no credential values or bearer URLs.
Harness / mode	Existing verify:concurrency, verify:api-keys, verify:privacy, verify:email, source tests; controlled egress check

KLM-D2-SEC-005 / Frame-origin and message spoofing

P0 | Security | Not Run | Issues: #114

Preconditions	Disposable accounts/fixtures only; local authorized fault injection; owner-approved egress test target for DNS case.
Steps	Send bridge replies from wrong origin/window with wrong ID; trusted parent attempts changed XDR.
Expected	Ignore untrusted replies and refuse tampered trusted replies; no wildcard origins, no broad frame access.
Required evidence	Sanitized request/response and DB evidence; guard test logs; no credential values or bearer URLs.
Harness / mode	Existing verify:concurrency, verify:api-keys, verify:privacy, verify:email, source tests; controlled egress check

KLM-D2-SEC-006 / Session cookie and CSRF boundaries

P0 | Security | Not Run | Issues: #114

Preconditions	Disposable accounts/fixtures only; local authorized fault injection; owner-approved egress test target for DNS case.
Steps	Enable exact Lantern origin; login in trusted partition; issue cookie mutation from foreign/originless browser; log out.
Expected	HttpOnly/Secure/Partitioned cookie matches deployment; unsafe browser mutations require exact Kleym origin; logout revokes/clears matching partition.
Required evidence	Sanitized request/response and DB evidence; guard test logs; no credential values or bearer URLs.
Harness / mode	Existing verify:concurrency, verify:api-keys, verify:privacy, verify:email, source tests; controlled egress check

KLM-D2-SEC-007 / Outbound URL canonicalization and redirects

P0 | Security | Not Run | Issues: #107

Preconditions	Disposable accounts/fixtures only; local authorized fault injection; owner-approved egress test target for DNS case.
Steps	Configure webhook/screening/OIDC/federation URL fixtures with local/private/reserved hosts, credentials, alternative numeric IPs, backslashes and 3xx.
Expected	Public HTTPS guard rejects unsafe forms and redirects; permitted loopback only in explicit development; network errors are safe and bounded.
Required evidence	Sanitized request/response and DB evidence; guard test logs; no credential values or bearer URLs.
Harness / mode	Existing verify:concurrency, verify:api-keys, verify:privacy, verify:email, source tests; controlled egress check

KLM-D2-SEC-008 / DNS rebinding and deployment egress

P0 | Security | Not Run | Issues: #107

Preconditions	Disposable accounts/fixtures only; local authorized fault injection; owner-approved egress test target for DNS case.
Steps	In controlled deployment resolve a public-looking allowed hostname to private IP; test socket egress policy with security owner.
Expected	No connection to private/internal address; DNS/socket or platform egress enforcement must prove this. Current URL-only guards do not satisfy this case.
Required evidence	Sanitized request/response and DB evidence; guard test logs; no credential values or bearer URLs.
Harness / mode	Existing verify:concurrency, verify:api-keys, verify:privacy, verify:email, source tests; controlled egress check

KLM-D2-SEC-009 / Audit seed emitted log and future key guard

P0 | Security | Not Run | Issues: #109

Preconditions	Disposable accounts/fixtures only; local authorized fault injection; owner-approved egress test target for DNS case.
Steps	Run secret-classification focused tests and guard using planted root/nested fields; add unclassified seed fixture; verify fixed public signature.
Expected	Audit/sponsor/encryption/provider fields censored; future seed name fails classification; safe public IDs observable; signature bytes/verification format unchanged.
Required evidence	Sanitized request/response and DB evidence; guard test logs; no credential values or bearer URLs.
Harness / mode	Existing verify:concurrency, verify:api-keys, verify:privacy, verify:email, source tests; controlled egress check

KLM-D2-SEC-010 / Client bundle and evidence leak scan

P0 | Security | Not Run | Issues: #108,#109

Preconditions	Disposable accounts/fixtures only; local authorized fault injection; owner-approved egress test target for DNS case.
Steps	Scan candidate source/build and proposed report text for seeds, mnemonic fixtures, bearer URLs, tokens and credentials; inspect all intended screenshots.
Expected	No real secret or complete claim fragment is published; public IDs/hashes permitted; scan scope/limitations recorded, not called security assurance.
Required evidence	Sanitized request/response and DB evidence; guard test logs; no credential values or bearer URLs.
Harness / mode	Existing verify:concurrency, verify:api-keys, verify:privacy, verify:email, source tests; controlled egress check

KLM-D2-SEC-011 / Bearer-email exclusion for pilot

P0 | Security | Not Run | Issues: #108

Preconditions	Disposable accounts/fixtures only; local authorized fault injection; owner-approved egress test target for DNS case.
Steps	Enable campaign/self-delivery mode; invoke legacy email/reminder queue path and inspect sanitized message metadata.
Expected	Pilot cannot enqueue/send full bearer links via built-in email; server never receives browser link key. Legacy behavior requires explicit gated product decision.
Required evidence	Sanitized request/response and DB evidence; guard test logs; no credential values or bearer URLs.
Harness / mode	Existing verify:concurrency, verify:api-keys, verify:privacy, verify:email, source tests; controlled egress check

KLM-D2-AUTH-001 / Fresh-auth and MFA protected operations

P0 | Security | Not Run | Issues: #109,#111,#114

Preconditions	Isolated migrated DB/Redis; disposable authentication configuration and public verification fixtures.
Steps	With disposable sender/admin, exercise protected operations with missing/expired fresh auth; complete TOTP/WebAuthn, revoke session and retry.
Expected	Required second factor and fresh-auth gate cannot be bypassed; no protected mutation with revoked session; recovery material never enters evidence.
Required evidence	Sanitized authorization result, public export/signature and verifier output; no MFA codes/keys.
Harness / mode	verify:mfa; verify:sso; verify:approvals; verify:compliance; verify:receipts; verify:audit-chain

KLM-D2-AUTH-002 / Approval / screening hold and outage

P0 | Security | Not Run | Issues: #109,#111,#114

Preconditions	Isolated migrated DB/Redis; disposable authentication configuration and public verification fixtures.
Steps	Create approval-required and screened campaign fixtures; attempt funding before approval, on explicit hold and while screening dependency is unreachable.
Expected	Authorization and hold gates enforced; unavailable screening fails closed with safe actionable result; only authorized approved campaign can advance.
Required evidence	Sanitized authorization result, public export/signature and verifier output; no MFA codes/keys.
Harness / mode	verify:mfa; verify:sso; verify:approvals; verify:compliance; verify:receipts; verify:audit-chain

KLM-D2-AUTH-003 / Signed export tampering and key history

P0 | Security | Not Run | Issues: #109,#111,#114

Preconditions	Isolated migrated DB/Redis; disposable authentication configuration and public verification fixtures.
Steps	Generate receipts/audit export with disposable signing configuration; verify publicly; alter row/amount/signature and verify again; verify archived public-key fixture.
Expected	Valid evidence verifies without a signing seed; tamper fails; canonical/version/key history preserved; private credentials and bearer metadata absent.
Required evidence	Sanitized authorization result, public export/signature and verifier output; no MFA codes/keys.
Harness / mode	verify:mfa; verify:sso; verify:approvals; verify:compliance; verify:receipts; verify:audit-chain

D3 / Desktop, Android and Lantern wallet journeys

Every case below starts Not Run. Record exact candidate, observed result and evidence before changing status.

KLM-D3-FRE-001 / Desktop Freighter sender connection

P0 | UAT | Not Run | Issues: #97,#106,#114

Preconditions	G0-G3 gates passed; disposable Testnet identities; fresh campaign; privately backed-up keys. Exact browser, extension/APK, Android and Freighter versions recorded; approved public Testnet candidate.
Steps	Cold-open approved HTTPS sender page with unlocked desktop Freighter/Testnet. Connect intended public sender and prepare one funding batch.
Expected	Desktop extension control only; no Android app button; exact address/network visible; connection alone moves no funds.
Required evidence	Physical-device/desktop screenshots with private link hidden; recorded approvals, public hashes and independent balances.
Harness / mode	Manual actual desktop Freighter and Android Chrome + Freighter; screenshot automation cannot approve signatures

KLM-D3-FRE-002 / Desktop Freighter funding approval

P0 | UAT | Not Run | Issues: #97,#106,#114

Preconditions	G0-G3 gates passed; disposable Testnet identities; fresh campaign; privately backed-up keys. Exact browser, extension/APK, Android and Freighter versions recorded; approved public Testnet candidate.
Steps	Save private export; sign intended batch; inspect popup and submit; wait for campaign ledger confirmation.
Expected	Popup appears; correct official issuer/principal/signers; sender XLM warning, if wallet displays it, does not substitute for sponsor-paid ledger proof.
Required evidence	Physical-device/desktop screenshots with private link hidden; recorded approvals, public hashes and independent balances.
Harness / mode	Manual actual desktop Freighter and Android Chrome + Freighter; screenshot automation cannot approve signatures

KLM-D3-FRE-003 / Desktop recipient claim

P0 | UAT | Not Run | Issues: #97,#106,#114

Preconditions	G0-G3 gates passed; disposable Testnet identities; fresh campaign; privately backed-up keys. Exact browser, extension/APK, Android and Freighter versions recorded; approved public Testnet candidate.
Steps	Open link A privately in recipient browser; connect intended recipient; click Claim; review signatures and wait for settlement.
Expected	Exact one USDC credited once; confirmed transaction link uses Testnet; cleanup/reserves satisfy D1 accounting.
Required evidence	Physical-device/desktop screenshots with private link hidden; recorded approvals, public hashes and independent balances.
Harness / mode	Manual actual desktop Freighter and Android Chrome + Freighter; screenshot automation cannot approve signatures

KLM-D3-FRE-004 / Stale extension tab recovery

P0 | UAT | Not Run | Issues: #97,#106,#114

Preconditions	G0-G3 gates passed; disposable Testnet identities; fresh campaign; privately backed-up keys. Exact browser, extension/APK, Android and Freighter versions recorded; approved public Testnet candidate.
Steps	Keep saved draft, invalidate/restart extension bridge, compare original/fresh tabs, reopen only after private backup.
Expected	No endless spinner or false funded state; helpful reconnect guidance; new tab can recover public draft and keys only from user-held backup.
Required evidence	Physical-device/desktop screenshots with private link hidden; recorded approvals, public hashes and independent balances.
Harness / mode	Manual actual desktop Freighter and Android Chrome + Freighter; screenshot automation cannot approve signatures

KLM-D3-FRE-005 / Android Chrome phone controls

P0 | UAT | Not Run | Issues: #97,#106,#114

Preconditions	G0-G3 gates passed; disposable Testnet identities; fresh campaign; privately backed-up keys. Exact browser, extension/APK, Android and Freighter versions recorded; approved public Testnet candidate.
Steps	Open link B privately in Android Chrome without desktop extension. Inspect wallet controls and press phone connect.
Expected	No desktop-specific Connect wallet button. Freighter phone open/pair action is clear; connection is distinct from claim; installed/uninstalled fallback is truthful.
Required evidence	Physical-device/desktop screenshots with private link hidden; recorded approvals, public hashes and independent balances.
Harness / mode	Manual actual desktop Freighter and Android Chrome + Freighter; screenshot automation cannot approve signatures

KLM-D3-FRE-006 / Android Freighter connection return

P0 | UAT | Not Run | Issues: #97,#106,#114

Preconditions	G0-G3 gates passed; disposable Testnet identities; fresh campaign; privately backed-up keys. Exact browser, extension/APK, Android and Freighter versions recorded; approved public Testnet candidate.
Steps	Approve connection in installed Android Freighter, return to original Chrome tab and inspect account/network.
Expected	Tab reaches explicit Claim control with intended public account; connection alone cannot transfer; return guidance explains next step.
Required evidence	Physical-device/desktop screenshots with private link hidden; recorded approvals, public hashes and independent balances.
Harness / mode	Manual actual desktop Freighter and Android Chrome + Freighter; screenshot automation cannot approve signatures

KLM-D3-FRE-007 / Android Freighter full claim signing

P0 | UAT | Not Run | Issues: #97,#106,#114

Preconditions	G0-G3 gates passed; disposable Testnet identities; fresh campaign; privately backed-up keys. Exact browser, extension/APK, Android and Freighter versions recorded; approved public Testnet candidate.
Steps	From connected link B in Chrome press Claim; open wallet review, explicitly approve, return and confirm independently.
Expected	Actual v1.25.28 or recorded installed-version mobile signatures work for required operations; exact one USDC; no manual raw-XDR or key entry.
Required evidence	Physical-device/desktop screenshots with private link hidden; recorded approvals, public hashes and independent balances.
Harness / mode	Manual actual desktop Freighter and Android Chrome + Freighter; screenshot automation cannot approve signatures

KLM-D3-FRE-008 / Android reconnect/background lifecycle

P0 | UAT | Not Run | Issues: #97,#106,#114

Preconditions	G0-G3 gates passed; disposable Testnet identities; fresh campaign; privately backed-up keys. Exact browser, extension/APK, Android and Freighter versions recorded; approved public Testnet candidate.
Steps	Background Chrome/Freighter during connection, review and submission; resume; test expiry and user rejection separately.
Expected	No duplicate submission on resume; usable safe recovery; stale wallet context cannot sign a changed transaction; keys remain private.
Required evidence	Physical-device/desktop screenshots with private link hidden; recorded approvals, public hashes and independent balances.
Harness / mode	Manual actual desktop Freighter and Android Chrome + Freighter; screenshot automation cannot approve signatures

KLM-D3-FRE-009 / App not installed / pairing expiry

P0 | UAT | Not Run | Issues: #97,#106,#114

Preconditions	G0-G3 gates passed; disposable Testnet identities; fresh campaign; privately backed-up keys. Exact browser, extension/APK, Android and Freighter versions recorded; approved public Testnet candidate.
Steps	Use Android without installed app or expired WalletConnect pairing; press connect and cancel/retry.
Expected	Usable fallback/help and bounded timeout; no claim or phantom connected state; no promise that app opening equals approval.
Required evidence	Physical-device/desktop screenshots with private link hidden; recorded approvals, public hashes and independent balances.
Harness / mode	Manual actual desktop Freighter and Android Chrome + Freighter; screenshot automation cannot approve signatures

KLM-D3-LAN-001 / Curated directory entry and package identity

P0 | UAT | Not Run | Issues: #106,#114

Preconditions	G0-G3 gates passed; disposable Testnet identities; fresh campaign; privately backed-up keys. G3-Lantern: rebuilt curated package, verified origins and HTTPS partition-cookie support required.
Steps	Hash senior-provided Lantern APK/extension; inspect Apps directory for exact approved Kleym HTTPS origin/session=true; compare effective installed extension origin.
Expected	Kleym is explicitly curated; Android directory metadata generated; exact approved frame origins. Current 0.6.2 without Kleym cannot pass this prerequisite.
Required evidence	Package SHA256, exact origin/config record, sanitized bridge type/ID trace, physical wallet review and public ledger proof.
Harness / mode	Manual rebuilt Lantern extension/APK; tests/lib/wallet/lantern*.test.ts and verify-lantern-browser are supplementary fixtures

KLM-D3-LAN-002 / Android public sender handoff

P0 | UAT | Not Run | Issues: #106,#114

Preconditions	G0-G3 gates passed; disposable Testnet identities; fresh campaign; privately backed-up keys. G3-Lantern: rebuilt curated package, verified origins and HTTPS partition-cookie support required.
Steps	From public Kleym sender/login in Chrome select Open in Lantern; inspect opened page and browser fallback with app absent.
Expected	Only public path handed off; query/fragment removed; app opens curated Kleym, fallback is intended public page; no private claim fragment copied into intent.
Required evidence	Package SHA256, exact origin/config record, sanitized bridge type/ID trace, physical wallet review and public ledger proof.
Harness / mode	Manual rebuilt Lantern extension/APK; tests/lib/wallet/lantern*.test.ts and verify-lantern-browser are supplementary fixtures

KLM-D3-LAN-003 / Lantern sender authenticated session

P0 | UAT | Not Run | Issues: #106,#114

Preconditions	G0-G3 gates passed; disposable Testnet identities; fresh campaign; privately backed-up keys. G3-Lantern: rebuilt curated package, verified origins and HTTPS partition-cookie support required.
Steps	Inside curated trusted Lantern frame login, connect wallet, create draft, leave/reopen Apps and log out.
Expected	Authenticated sender experience works in correct cookie partition; desktop browser session is separate; logout terminates partition session.
Required evidence	Package SHA256, exact origin/config record, sanitized bridge type/ID trace, physical wallet review and public ledger proof.
Harness / mode	Manual rebuilt Lantern extension/APK; tests/lib/wallet/lantern*.test.ts and verify-lantern-browser are supplementary fixtures

KLM-D3-LAN-004 / Lantern desktop funding and management

P0 | UAT | Not Run | Issues: #106,#114

Preconditions	G0-G3 gates passed; disposable Testnet identities; fresh campaign; privately backed-up keys. G3-Lantern: rebuilt curated package, verified origins and HTTPS partition-cookie support required.
Steps	In rebuilt extension open Apps > Kleym. Create saved 3-link campaign; approve funding; inspect campaign and reclaim unused link after expiry.
Expected	Lantern receives exact sign-only XDR; no external Freighter phone control; API submits and confirms exactly once; auth/organization management remains intact.
Required evidence	Package SHA256, exact origin/config record, sanitized bridge type/ID trace, physical wallet review and public ledger proof.
Harness / mode	Manual rebuilt Lantern extension/APK; tests/lib/wallet/lantern*.test.ts and verify-lantern-browser are supplementary fixtures

KLM-D3-LAN-005 / Lantern Android funding and management

P0 | UAT | Not Run | Issues: #106,#114

Preconditions	G0-G3 gates passed; disposable Testnet identities; fresh campaign; privately backed-up keys. G3-Lantern: rebuilt curated package, verified origins and HTTPS partition-cookie support required.
Steps	Repeat saved 3-link campaign creation/funding in rebuilt APK; inspect scrolling, approve review and confirmed dashboard.
Expected	Full creator journey works on physical phone, not only mocked bridge; exact hash/account/network preserved; no overflow or inaccessible signing action.
Required evidence	Package SHA256, exact origin/config record, sanitized bridge type/ID trace, physical wallet review and public ledger proof.
Harness / mode	Manual rebuilt Lantern extension/APK; tests/lib/wallet/lantern*.test.ts and verify-lantern-browser are supplementary fixtures

KLM-D3-LAN-006 / Recipient claim already in trusted Lantern frame

P0 | UAT | Not Run | Issues: #106,#114

Preconditions	G0-G3 gates passed; disposable Testnet identities; fresh campaign; privately backed-up keys. G3-Lantern: rebuilt curated package, verified origins and HTTPS partition-cookie support required.
Steps	Open disposable link inside curated trusted frame using owner-approved private-link route; connect Lantern and explicitly claim.
Expected	Connect Lantern selected; no Freighter pairing control; exact claim review inside wallet; confirmed asset reaches selected recipient; public launcher is not claimed to move private links.
Required evidence	Package SHA256, exact origin/config record, sanitized bridge type/ID trace, physical wallet review and public ledger proof.
Harness / mode	Manual rebuilt Lantern extension/APK; tests/lib/wallet/lantern*.test.ts and verify-lantern-browser are supplementary fixtures

KLM-D3-LAN-007 / Uncurated generic URL sandbox

P0 | UAT | Not Run | Issues: #106,#114

Preconditions	G0-G3 gates passed; disposable Testnet identities; fresh campaign; privately backed-up keys. G3-Lantern: rebuilt curated package, verified origins and HTTPS partition-cookie support required.
Steps	Open Kleym via Lantern generic URL bar instead of approved Apps directory; attempt login/connect safely.
Expected	Opaque sandbox is not treated as trusted authenticated integration; honest guidance; no fallback widening origins or copying private fragments.
Required evidence	Package SHA256, exact origin/config record, sanitized bridge type/ID trace, physical wallet review and public ledger proof.
Harness / mode	Manual rebuilt Lantern extension/APK; tests/lib/wallet/lantern*.test.ts and verify-lantern-browser are supplementary fixtures

KLM-D3-LAN-008 / Lantern delayed idless connection reply

P0 | UAT | Not Run | Issues: #106,#114

Preconditions	G0-G3 gates passed; disposable Testnet identities; fresh campaign; privately backed-up keys. G3-Lantern: rebuilt curated package, verified origins and HTTPS partition-cookie support required.
Steps	Delay getPublicKey reply past timeout, retry without reopening, then deliver old response in bridge fixture.
Expected	Late response cannot complete a different connection; reopen requirement visible; signing still correlates exact request ID/source/origin.
Required evidence	Package SHA256, exact origin/config record, sanitized bridge type/ID trace, physical wallet review and public ledger proof.
Harness / mode	Manual rebuilt Lantern extension/APK; tests/lib/wallet/lantern*.test.ts and verify-lantern-browser are supplementary fixtures

KLM-D3-LAN-009 / Lantern wrong network / account / permissions

P0 | UAT | Not Run | Issues: #106,#114

Preconditions	G0-G3 gates passed; disposable Testnet identities; fresh campaign; privately backed-up keys. G3-Lantern: rebuilt curated package, verified origins and HTTPS partition-cookie support required.
Steps	Switch network/account or revoke app grant before sign; return wrong identity or tampered signature via fixture.
Expected	Fresh grant/account/network checks refuse; never signAndSubmit behind Kleym journal; original reviewed transaction unchanged.
Required evidence	Package SHA256, exact origin/config record, sanitized bridge type/ID trace, physical wallet review and public ledger proof.
Harness / mode	Manual rebuilt Lantern extension/APK; tests/lib/wallet/lantern*.test.ts and verify-lantern-browser are supplementary fixtures

KLM-D3-UX-001 / Keyboard and focus order

P1 | UAT | Not Run | Issues: #106,#114

Preconditions	Isolated UI candidate; disposable public snapshots; real assistive tools for accessibility assertions.
Steps	Use Tab/Shift-Tab/Enter/Space from login through amount editor, expiry picker, review, export and wallet controls.
Expected	All interactive elements reachable with visible focus; logical order; no keyboard trap.
Required evidence	Screenshots/notes with private areas excluded; tool/browser versions; elapsed journey and exact confusing text.
Harness / mode	Manual keyboard/NVDA/TalkBack; production Chromium layout fixtures supplementary

KLM-D3-UX-002 / TalkBack / NVDA status announcements

P1 | UAT | Not Run | Issues: #106,#114

Preconditions	Isolated UI candidate; disposable public snapshots; real assistive tools for accessibility assertions.
Steps	Use Android TalkBack and desktop NVDA for invalid amount, funding pending and claim result.
Expected	Errors and new states announced meaningfully; controls labelled; no reliance on colour alone.
Required evidence	Screenshots/notes with private areas excluded; tool/browser versions; elapsed journey and exact confusing text.
Harness / mode	Manual keyboard/NVDA/TalkBack; production Chromium layout fixtures supplementary

KLM-D3-UX-003 / Phone viewport and 200% zoom

P1 | UAT | Not Run | Issues: #106,#114

Preconditions	Isolated UI candidate; disposable public snapshots; real assistive tools for accessibility assertions.
Steps	Test 360px/393px portrait, landscape, keyboard-open state and desktop 200% zoom across creation/review/claim.
Expected	No sideways scrolling or clipped primary controls; amount/date input and signing return usable.
Required evidence	Screenshots/notes with private areas excluded; tool/browser versions; elapsed journey and exact confusing text.
Harness / mode	Manual keyboard/NVDA/TalkBack; production Chromium layout fixtures supplementary

KLM-D3-UX-004 / Cold recipient without team help

P1 | UAT | Not Run | Issues: #106,#114

Preconditions	Isolated UI candidate; disposable public snapshots; real assistive tools for accessibility assertions.
Steps	Give independent reviewer only their private test link and minimal Testnet warning; time connection/claim and record hesitation.
Expected	Reviewer identifies asset/network, connection versus claim and result without keys/raw-XDR instruction; friction recorded as UAT findings, not fabricated passes.
Required evidence	Screenshots/notes with private areas excluded; tool/browser versions; elapsed journey and exact confusing text.
Harness / mode	Manual keyboard/NVDA/TalkBack; production Chromium layout fixtures supplementary

KLM-D3-ADM-001 / Campaign list, filters and per-link history

P1 | Regression | Not Run | Issues: #100,#106,#114

Preconditions	Isolated UI/API fixtures or approved public candidate; private claim URL never captured.
Steps	Create mixed-status equal/custom campaigns in two orgs; filter/search/page dashboard and campaign details; refresh after settlement.
Expected	No cross-org data; stable paging/counts; exact per-link amount and confirmed lifecycle status; no stale active totals from submitted-only attempts.
Required evidence	Scoped API snapshots, dashboard screenshots and safe state fixture output.
Harness / mode	verify:campaign-list; verify:campaign-detail; verify:templates; e2e/claim-page.spec.ts

KLM-D3-ADM-002 / Templates and historical funding compatibility

P1 | Regression | Not Run | Issues: #100,#106,#114

Preconditions	Isolated UI/API fixtures or approved public candidate; private claim URL never captured.
Steps	Create draft from existing template and inspect funding mode; compare with new sponsored API campaign and historical sender-paid campaign.
Expected	Template/historical mode remains explicit and supported; no silent migration to sponsored version; funding signatures/costs match actual reviewed protocol.
Required evidence	Scoped API snapshots, dashboard screenshots and safe state fixture output.
Harness / mode	verify:campaign-list; verify:campaign-detail; verify:templates; e2e/claim-page.spec.ts

KLM-D3-ADM-003 / Claimed / expired / invalid link copy

P1 | Regression | Not Run | Issues: #100,#106,#114

Preconditions	Isolated UI/API fixtures or approved public candidate; private claim URL never captured.
Steps	Open safe local fixtures for already claimed, expired-unclaimed, invalid/missing fragment and provider outage.
Expected	Correct distinct states and actionable messages; no false funds-in-wallet message before confirmation; no spinner stuck or secret-bearing error.
Required evidence	Scoped API snapshots, dashboard screenshots and safe state fixture output.
Harness / mode	verify:campaign-list; verify:campaign-detail; verify:templates; e2e/claim-page.spec.ts

D4 / Public candidate, operations and release qualification

Every case below starts Not Run. Record exact candidate, observed result and evidence before changing status.

KLM-D4-OPS-001 / Candidate freeze and deployed parity

P0 | UAT | Not Run | Issues: #111,#114

Preconditions	G0 candidate recorded; isolated infrastructure available; deployment and Mainnet changes require separate approval.
Steps	Record feature commit, dependency lock hash, migrations, runtime/package-manager versions, effective public URL and server build identity; compare release manifest; run dependency audit with supported package manager.
Expected	Tested source, built bytes and deployed candidate match; unknown identity blocks qualification. Unresolved critical/high dependency advisories block release until remediated or explicitly assessed by the security owner.
Required evidence	Release manifest, sanitized logs/runbook observations, owner decisions, metrics/restore notes and independent public ledger data where authorized.
Harness / mode	Existing verification scripts and deployment/runbook procedures; no automatic production writes

KLM-D4-OPS-002 / Migration and rollback on populated copy

P0 | UAT | Not Run | Issues: #98,#115

Preconditions	G0 candidate recorded; isolated infrastructure available; deployment and Mainnet changes require separate approval.
Steps	Backup isolated seeded DB with campaigns/attempts/receipts; apply candidate migration; exercise old/new states; rehearse documented rollback from copy.
Expected	No precision/history/lineage loss; safe recovery of pending attempts; rollback decision tested and source/deploy artifact preserved.
Required evidence	Release manifest, sanitized logs/runbook observations, owner decisions, metrics/restore notes and independent public ledger data where authorized.
Harness / mode	Existing verification scripts and deployment/runbook procedures; no automatic production writes

KLM-D4-OPS-003 / Real DB/Redis CI qualification

P0 | UAT | Not Run | Issues: #111,#112,#114

Preconditions	G0 candidate recorded; isolated infrastructure available; deployment and Mainnet changes require separate approval.
Steps	Run full isolated PostgreSQL/Redis suites from candidate in CI; include keys, receipts, batches, claims, source coordination, backup and recovery.
Expected	Green evidence for candidate includes meaningful assertions; unavailable Docker/local services mean Blocked for execution, not pass from mocks.
Required evidence	Release manifest, sanitized logs/runbook observations, owner decisions, metrics/restore notes and independent public ledger data where authorized.
Harness / mode	Existing verification scripts and deployment/runbook procedures; no automatic production writes

KLM-D4-OPS-004 / Public HTTPS cold sender and recipient

P0 | UAT | Not Run | Issues: #114

Preconditions	G0 candidate recorded; isolated infrastructure available; deployment and Mainnet changes require separate approval.
Steps	After owner-approved deployment, cold-open public URL on desktop and Android outside team network; execute 3-link tutorial.
Expected	HTTPS reachable, Testnet clearly labelled; no localhost dependency; account/claim links work without team browser profile.
Required evidence	Release manifest, sanitized logs/runbook observations, owner decisions, metrics/restore notes and independent public ledger data where authorized.
Harness / mode	Existing verification scripts and deployment/runbook procedures; no automatic production writes

KLM-D4-OPS-005 / 200-link cost/capacity budget

P0 | UAT | Not Run | Issues: #99,#103,#115

Preconditions	G0 candidate recorded; isolated infrastructure available; deployment and Mainnet changes require separate approval.
Steps	Record fresh base reserve, fees, sponsor liabilities and configured campaign/recipient caps; quote 200 one-USDC links then enforce limiting budgets.
Expected	Principal separate from operational XLM; fee bid versus charged fee distinguished; persistent recipient reserves included in lifecycle budget; configured limits enforced before spend.
Required evidence	Release manifest, sanitized logs/runbook observations, owner decisions, metrics/restore notes and independent public ledger data where authorized.
Harness / mode	Existing verification scripts and deployment/runbook procedures; no automatic production writes

KLM-D4-OPS-006 / Performance latency/provider-read budget

P0 | Load | Not Run | Issues: #103,#114

Preconditions	G0 candidate recorded; isolated infrastructure available; deployment and Mainnet changes require separate approval.
Steps	Measure 30 representative prepare/resolve requests at agreed concurrency with recorded network and fixture dataset; report median/p95, provider calls and errors.
Expected	Baseline and candidate measured comparably; no invented speedup; static 10-to-5 provider call reduction is not latency evidence. Owner sets SLO before sign-off.
Required evidence	Release manifest, sanitized logs/runbook observations, owner decisions, metrics/restore notes and independent public ledger data where authorized.
Harness / mode	Existing verification scripts and deployment/runbook procedures; no automatic production writes

KLM-D4-OPS-007 / API throttling / provider rate limits

P0 | Load | Not Run | Issues: #104,#113

Preconditions	G0 candidate recorded; isolated infrastructure available; deployment and Mainnet changes require separate approval.
Steps	Use bounded local load within agreed test budget; return 429/503/timeouts from isolated provider; observe recovery and queues.
Expected	Useful bounded response/backoff; no storm, duplicate spend or false active state; rate limit shown honestly and financial journal retained.
Required evidence	Release manifest, sanitized logs/runbook observations, owner decisions, metrics/restore notes and independent public ledger data where authorized.
Harness / mode	Existing verification scripts and deployment/runbook procedures; no automatic production writes

KLM-D4-OPS-008 / Sponsor failover / shortage / refill

P0 | UAT | Not Run | Issues: #103,#113,#115

Preconditions	G0 candidate recorded; isolated infrastructure available; deployment and Mainnet changes require separate approval.
Steps	Exhaust primary fixture below working threshold; select secondary; test total pool shortage and alert routing; rehearse refill procedure without cold signing.
Expected	Fresh sponsor choice respects outstanding liability and pinned campaign sponsor; alerts actionable; no unauthorized automatic cold refill; approved owners recorded.
Required evidence	Release manifest, sanitized logs/runbook observations, owner decisions, metrics/restore notes and independent public ledger data where authorized.
Harness / mode	Existing verification scripts and deployment/runbook procedures; no automatic production writes

KLM-D4-OPS-009 / Worker pause and drain

P0 | UAT | Not Run | Issues: #104,#113

Preconditions	G0 candidate recorded; isolated infrastructure available; deployment and Mainnet changes require separate approval.
Steps	Pause creation with pending submitted attempts; drain/reconcile workers before planned shutdown, then restart isolated services.
Expected	User-facing pause clear; accepted attempts remain durable and settle; no stranded state hidden by queue deletion.
Required evidence	Release manifest, sanitized logs/runbook observations, owner decisions, metrics/restore notes and independent public ledger data where authorized.
Harness / mode	Existing verification scripts and deployment/runbook procedures; no automatic production writes

KLM-D4-OPS-010 / Backup restore and disaster recovery

P0 | UAT | Not Run | Issues: #113,#115

Preconditions	G0 candidate recorded; isolated infrastructure available; deployment and Mainnet changes require separate approval.
Steps	Restore encrypted DB backup to isolated environment; restore matching configuration via secret store; reconcile exact pending hashes before unpausing.
Expected	Audit/history and financial journals preserved; no key in reports; measured recovery time/loss against owner-approved RTO/RPO.
Required evidence	Release manifest, sanitized logs/runbook observations, owner decisions, metrics/restore notes and independent public ledger data where authorized.
Harness / mode	Existing verification scripts and deployment/runbook procedures; no automatic production writes

KLM-D4-OPS-011 / Monitoring and signed evidence cadence

P0 | UAT | Not Run | Issues: #111,#113

Preconditions	G0 candidate recorded; isolated infrastructure available; deployment and Mainnet changes require separate approval.
Steps	Trigger stale pending attempt, low spendable reserve, queue delay and mismatch fixture; inspect dashboards/alerts and generate verified export.
Expected	Owner can trace safe correlation ID/hash; alerts reach approved operator; signed receipts verify; production cadence/gates documented and actually wired.
Required evidence	Release manifest, sanitized logs/runbook observations, owner decisions, metrics/restore notes and independent public ledger data where authorized.
Harness / mode	Existing verification scripts and deployment/runbook procedures; no automatic production writes

KLM-D4-OPS-012 / Mainnet read-only config and issuer

P0 | UAT | Not Run | Issues: #115

Preconditions	G0 candidate recorded; isolated infrastructure available; deployment and Mainnet changes require separate approval.
Steps	Inspect public passphrase/official USDC issuer and exact public Horizon identity; verify release checklist without signing or submission.
Expected	Public issuer GA5ZSEJYB37JRC5AVCIA5MOP4RHTM335X2KGX3IHOJAPP5RE34K4KZVN matches configured asset; no Testnet address/passphrase/issuer crossover. No live movement authorized by this plan.
Required evidence	Release manifest, sanitized logs/runbook observations, owner decisions, metrics/restore notes and independent public ledger data where authorized.
Harness / mode	Existing verification scripts and deployment/runbook procedures; no automatic production writes

KLM-D4-OPS-013 / Capped Mainnet pilot approval

P0 | UAT | Not Run | Issues: #115

Preconditions	G0 candidate recorded; isolated infrastructure available; deployment and Mainnet changes require separate approval.
Steps	After mandatory Testnet/operations cases pass, obtain recorded owner-approved limits, sponsor capital, incident/refill owners and rollback criteria.
Expected	Separate explicit approval required before any real-USDC transaction. Unresolved P0/financial evidence blocks live pilot regardless of local green suite.
Required evidence	Release manifest, sanitized logs/runbook observations, owner decisions, metrics/restore notes and independent public ledger data where authorized.
Harness / mode	Existing verification scripts and deployment/runbook procedures; no automatic production writes

KLM-D4-OPS-014 / Issue closure and evidence consistency

P0 | UAT | Not Run | Issues: #95,#114

Preconditions	G0 candidate recorded; isolated infrastructure available; deployment and Mainnet changes require separate approval.
Steps	For each proposed closure map every criterion to candidate source and executed case log; verify tracker totals/evidence/defect references.
Expected	Close only satisfied criteria; keep missing device/DB/ledger/deployment proof open; roadmap reflects qualified scope rather than promised launch date.
Required evidence	Release manifest, sanitized logs/runbook observations, owner decisions, metrics/restore notes and independent public ledger data where authorized.
Harness / mode	Existing verification scripts and deployment/runbook procedures; no automatic production writes

KLM-D4-DOC-001 / Privacy deletion and organization isolation

P1 | Regression | Not Run | Issues: #105,#108,#111,#114

Preconditions	Isolated data and telemetry sinks; source candidate and API docs frozen.
Steps	With disposable org/test user, export personal data, request allowed erasure, and verify financial audit retention and unrelated org isolation.
Expected	Only permitted personal data returned/erased; immutable financial proof preserved under documented retention; keys/bearer URLs absent.
Required evidence	Sanitized export/schema/attributes output; docs-to-case mapping and reviewer notes.
Harness / mode	verify:privacy; verify:observability; source/docs review

KLM-D4-DOC-002 / Telemetry and error observability minimization

P1 | Regression | Not Run | Issues: #105,#108,#111,#114

Preconditions	Isolated data and telemetry sinks; source candidate and API docs frozen.
Steps	Trigger bounded funding/claim failures; inspect emitted logs and tracing attributes using planted secret/bearer fixtures.
Expected	Safe correlation IDs/hash/state observable; no mnemonic, seed, complete bearer URL, signed-body dump or private token retained in logs/traces.
Required evidence	Sanitized export/schema/attributes output; docs-to-case mapping and reviewer notes.
Harness / mode	verify:privacy; verify:observability; source/docs review

KLM-D4-DOC-003 / API documentation and support limits

P1 | Regression | Not Run | Issues: #105,#108,#111,#114

Preconditions	Isolated data and telemetry sinks; source candidate and API docs frozen.
Steps	Compare public write API/SDK examples, wallet support matrix, funding versions and recovery help with executed candidate cases.
Expected	Docs describe only proved support, explicit async confirmation and custom-amount renewal limitation; unsupported Android/Lantern/Mainnet claims withheld.
Required evidence	Sanitized export/schema/attributes output; docs-to-case mapping and reviewer notes.
Harness / mode	verify:privacy; verify:observability; source/docs review